



A Brief Introduction to Deepfake Attribution & Deepfake Model Recognition

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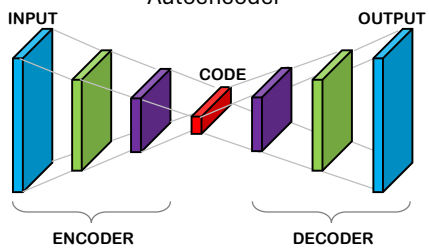
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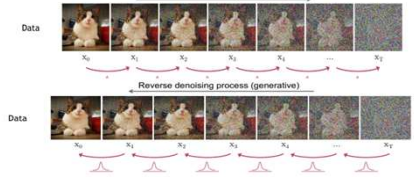
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Autoencoder Vs GAN Vs Diffusion Models

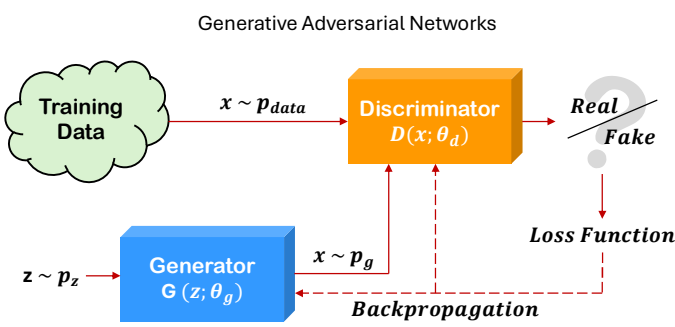
Autoencoder



Diffusion Models



Generative Adversarial Networks

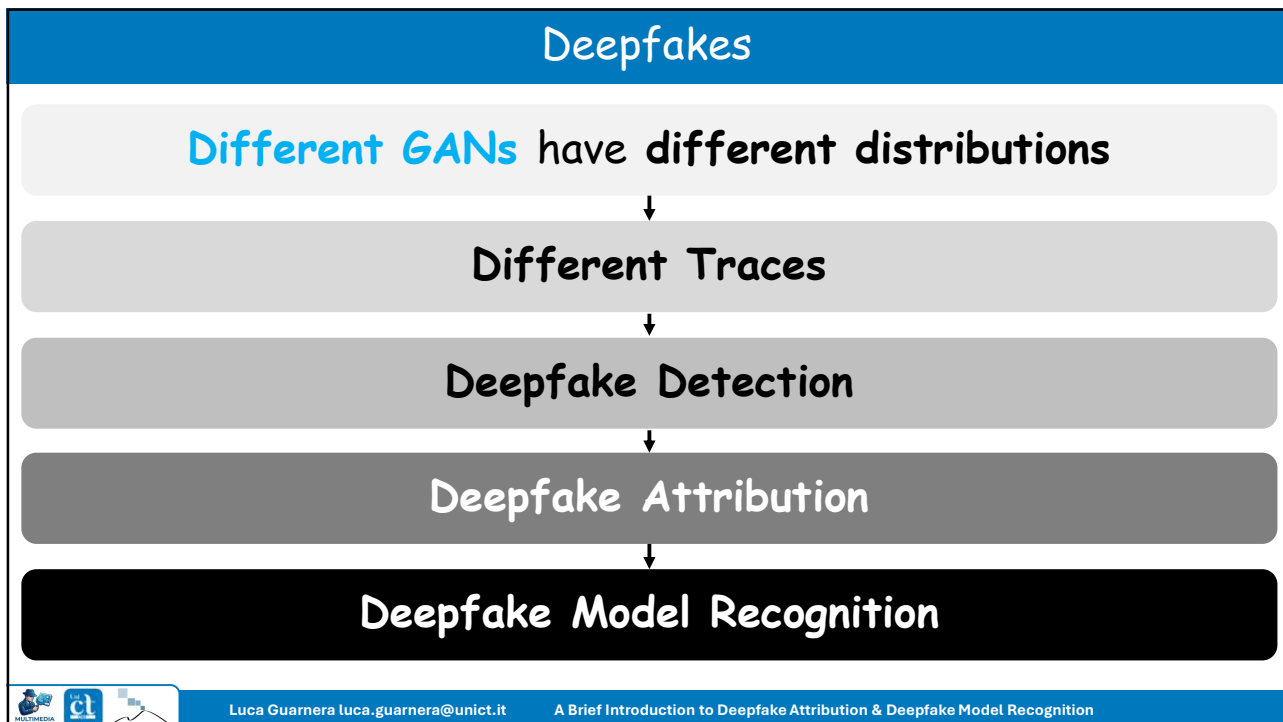




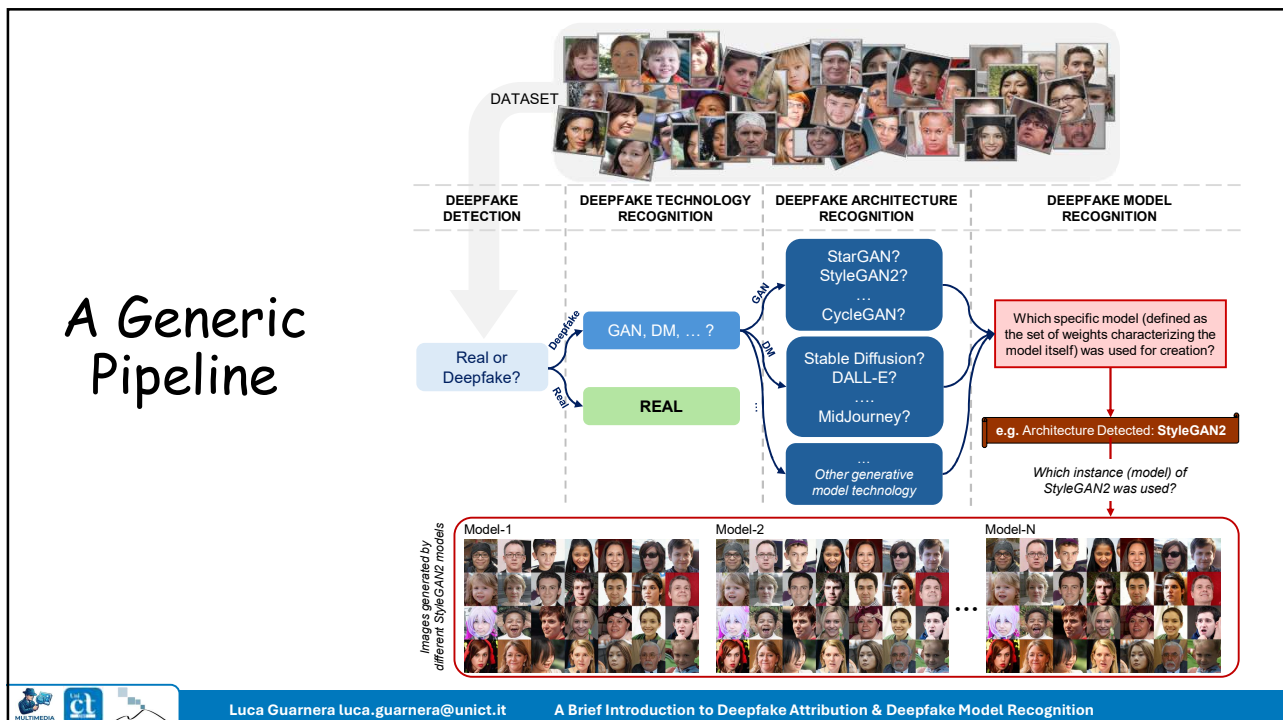
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2



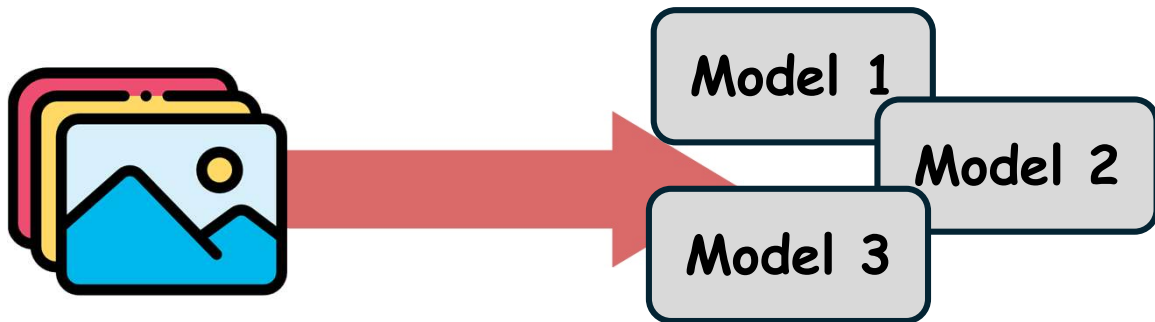
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Definizione

L'**attribuzione di deepfake** è il processo forense volto a identificare l'architettura generativa (es. GAN) o il modello specifico utilizzato per generare un contenuto sintetico (immagini, audio o video). È fondamentale per risalire all'origine della disinformazione e accertare le responsabilità legali o di sicurezza



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WILD: a new in-the-Wild Image Linkage Dataset for synthetic image attribution

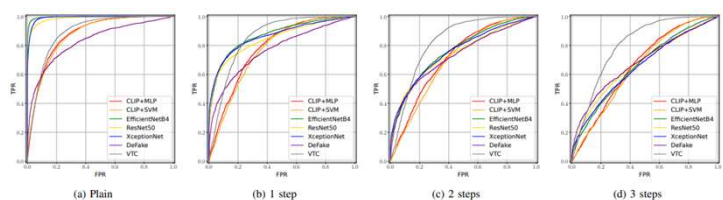
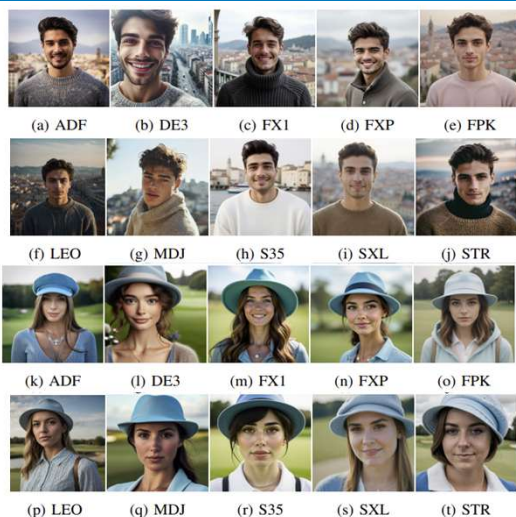


Fig. 3: ROC Curves of baselines in the binary classification task between open set (positive class) and closed set (negative task), with different levels of image post-processing: (a) plain images, (b) 1 post-processing step, (c) 2 post-processing steps, (d) 3 post-processing steps.

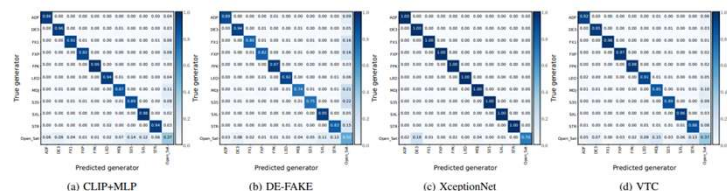


Fig. 4: Confusion matrices obtained by CLIP+MLP (a), DE-FAKE (b), XceptionNet (c), and VTC (d) on the open set attribution with rejection threshold, without post-processing. The generator-classes are represented by the three-letter codes defined in Section II-B.

Bongini, P., Mandelli, S., Montbeller, A., Casu, M., Pontorno, O., Ragaglia, C. V., ... & Barni, M. (2025, June). WILD: a new in-the-Wild Image Linkage Dataset for synthetic image attribution. In *2025 International Joint Conference on Neural Networks (IJCNN)* (pp. 1-8). IEEE.



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Le Sfide della Deepfake Attribution

L'attribuzione è molto più complessa della semplice *rilevazione* di un deepfake (cioè capire se un'immagine è falsa o meno):

- **Modelli in continua evoluzione:** Le tecnologie generative migliorano rapidamente, lasciando artefatti statistici sempre meno evidenti.
- **Sovrapposizione di strumenti:** Spesso i contenuti vengono manipolati usando più software diversi, rendendo difficile risalire a un singolo generatore.
- **Codice open-source:** La modifica e la redistribuzione di modelli di Intelligenza Artificiale open-source genera infinite varianti, confondendo le firme digitali



Come Funziona l'Analisi Forense

I ricercatori utilizzano metodologie specifiche per risalire all'origine del falso:

- **Analisi degli artefatti intrinseci:** Identificazione delle imperfezioni microscopiche e delle tracce uniche lasciate dai diversi modelli di IA.
- **Reti neurali dedicate:** Utilizzo di modelli basati sull'apprendimento automatico (come le reti CNN o ViT) addestrati a riconoscere la "mano" di specifici generatori.
- **Metodi «contrastivi»:** Tecniche avanzate che confrontano i falsi su larga scala per mantenere l'accuratezza dell'attribuzione anche su dataset di qualità inferiore o sconosciuti.



Closed-Set vs Open-Set Attribution

- Closed-set: all generators are known during training
- Open-set: unseen generators appear during testing
- Real-world forensic scenarios are naturally open-set
- Robust rejection mechanisms are essential



Main Challenges

- Generalization to unseen generators
- Compression and social-media processing
- Cross-dataset domain shifts
- Adversarial attacks and perturbations
- Similarities between diffusion pipelines



Deepfake Families

- GAN-based generation
- Diffusion-based generation
- Face swapping methods
- Fully synthetic video generation
- Hybrid and editing pipelines



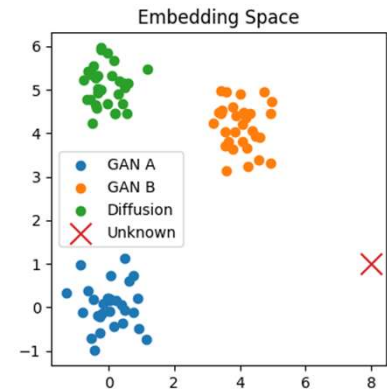
Fingerprint-Based Attribution

- Generator-specific artifacts
- Frequency-domain traces
- Upsampling inconsistencies
- Noise residuals and latent traces
- Decoder/VAE signatures



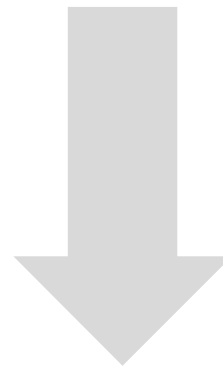
Metric Learning

- Learn embedding spaces for attribution
- Same-source samples are close together
- Different generators are pushed apart
- Triplet loss and contrastive learning are common



Hierarchical Attribution

- Step 1: Real vs Fake
- Step 2: GAN vs Diffusion
- Step 3: Specific source generator
- Unknown rejection at multiple levels



Ensemble Systems

- Multiple forensic experts combined
- Frequency analysis
- Spatial traces
- Semantic embeddings
- Voting and confidence fusion



Forensic Perspective

- Explainability is critical
- Attribution evidence must be reproducible
- Robustness against post-processing
- Chain-of-custody compatibility



Future Research Directions

- Video attribution
- Multimodal deepfake attribution
- Foundation-model fingerprints
- Adversarial robustness
- Real-world benchmark creation

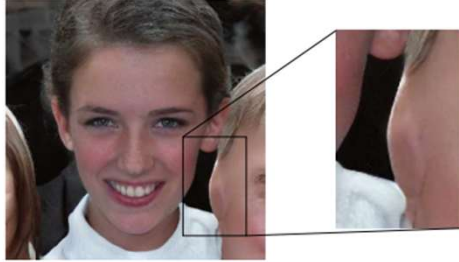


Summary

- Open-set attribution is a central forensic challenge
- Prototype and metric-learning methods are promising
- Robust rejection mechanisms are essential
- Future systems must generalize to unseen generators



On Attribution of Deepfakes



True model	Attributed model		
	ProgressiveGAN	StyleGAN	StyleGAN2
ProgressiveGAN	99.95%	0.05%	0.00%
StyleGAN	2.40%	94.00%	3.60%
StyleGAN2	0.85%	0.25%	98.90%

Zhang, B., Zhou, J. P., Shumailov, I., & Papernot, N. (2020). On attribution of deepfakes. *arXiv preprint arXiv:2008.09194*.



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Improving Generalization of Deepfake Detection by Training for Attribution

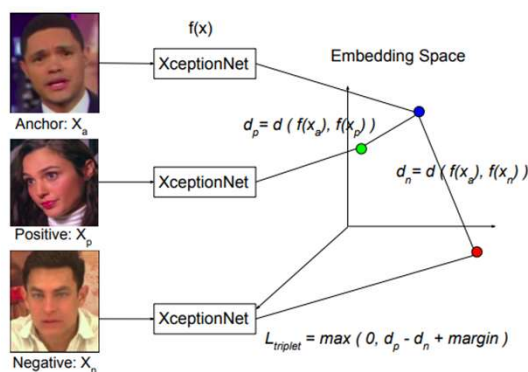


Fig. 3. Representation of the triplet loss network using the XceptionNet as the base model for generating embeddings of size 64.



Jain, A., Korshunov, P., & Marcel, S. (2021, October). Improving generalization of deepfake detection by training for attribution. In *2021 IEEE 23rd International Workshop on Multimedia Signal Processing (MMSP)* (pp. 1-6). IEEE.



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Mastering Deepfake Detection: A Cutting-Edge Approach to Distinguish GAN and Diffusion-Model Images

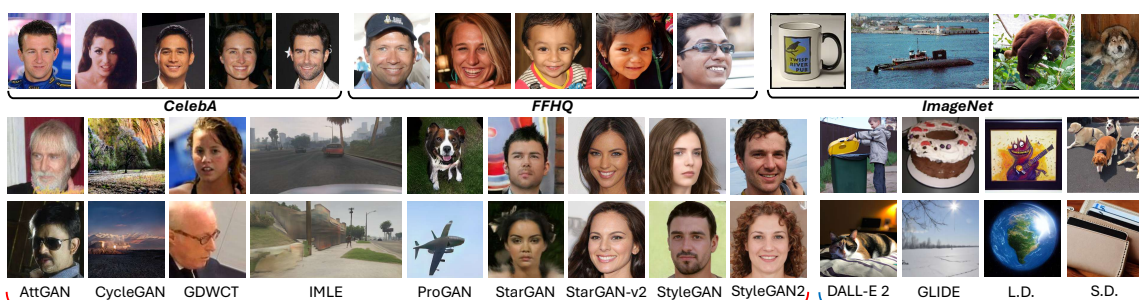
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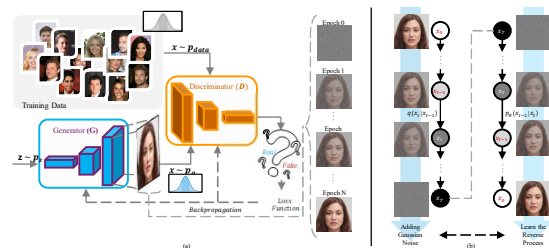
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Dataset



Classification Task	Train 50%	Val 20%	Test 30%
14-classes			
Total Images	28,000	4,200	7,000
#Img V class	2,000	300	500
13-classes			
Total Images	26,000	3,900	6,500
#Img V class	2,000	300	500
L1			
Total Images	46,480	11,620	24,900
#Img V class	23,240	5,810	12,450
L2			
Total Images	23,800	5,950	12,750
#Img V class	11,900	2,975	6,375
GANs			
Total Images	12,600	3,150	6,750
#Img V class	1,400	350	750
L3			
Total Images	11,200	2,800	6,000
#Img V class	2,800	700	1,500

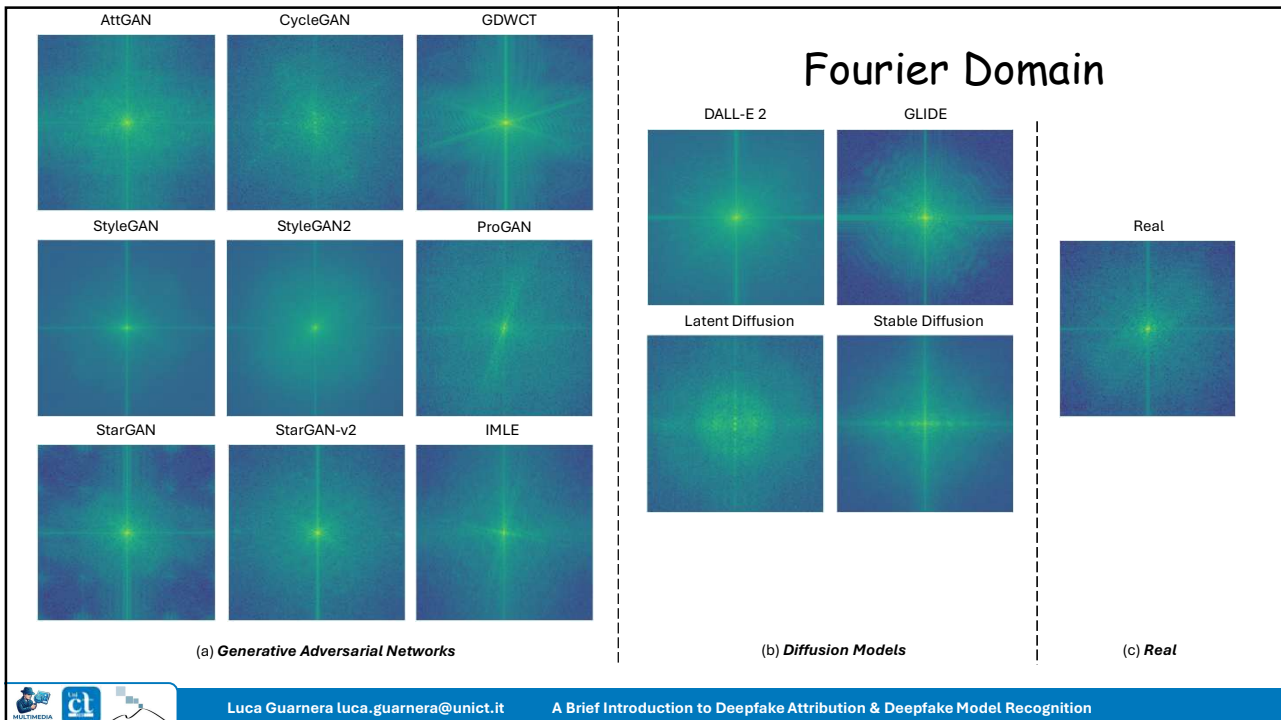
Table 1. Overview of the images employed for training, validation and test sets (last three columns with indication of % of samples). The first column denotes the classification task (e.g., 14 ~ classes is the flat classification task with 14 classes; L1 refers to the Level 1 of hierarchy). The *Total Images* rows indicate the total number of images employed for training, validation, and testing phases. The *#Img V class* represents the number of samples considered for each class.



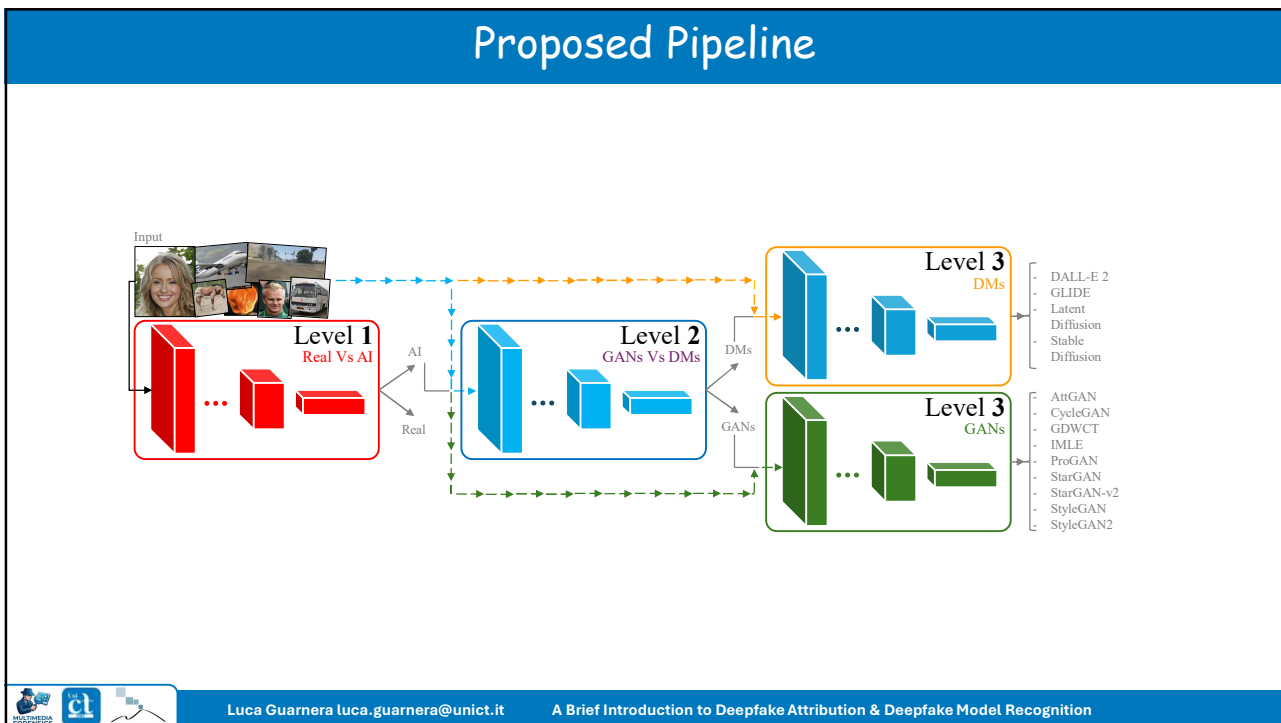
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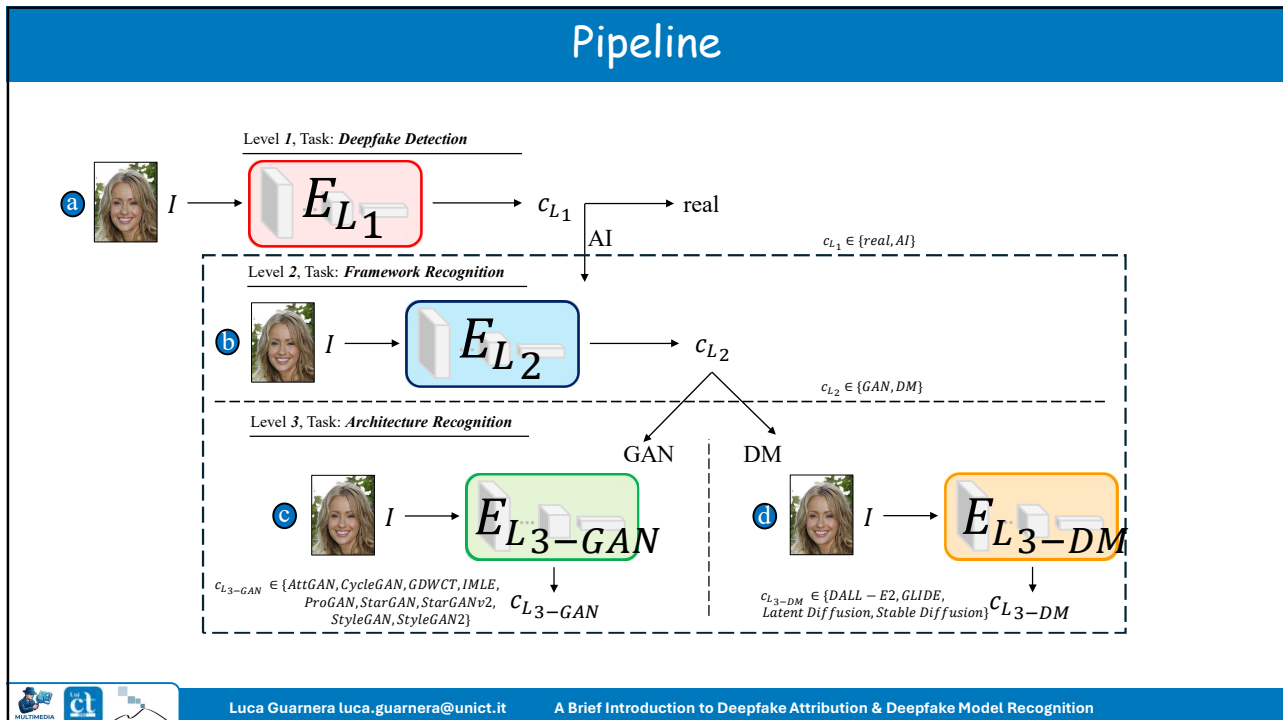
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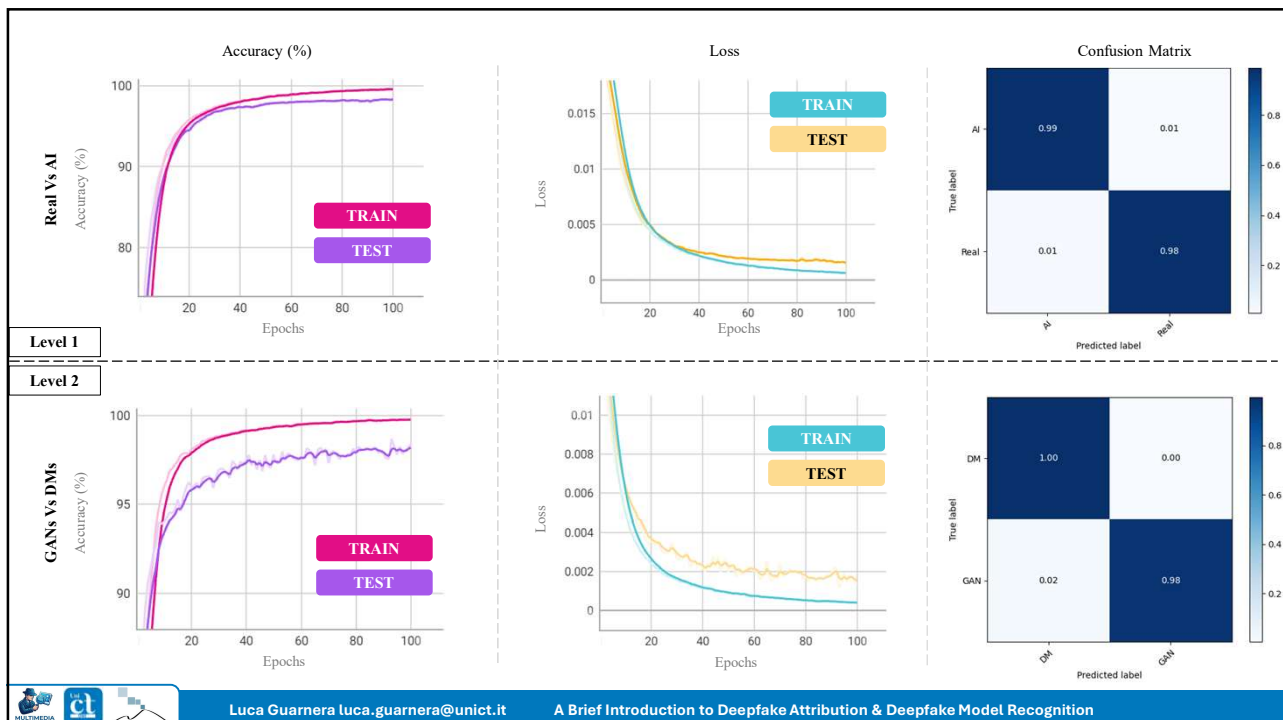
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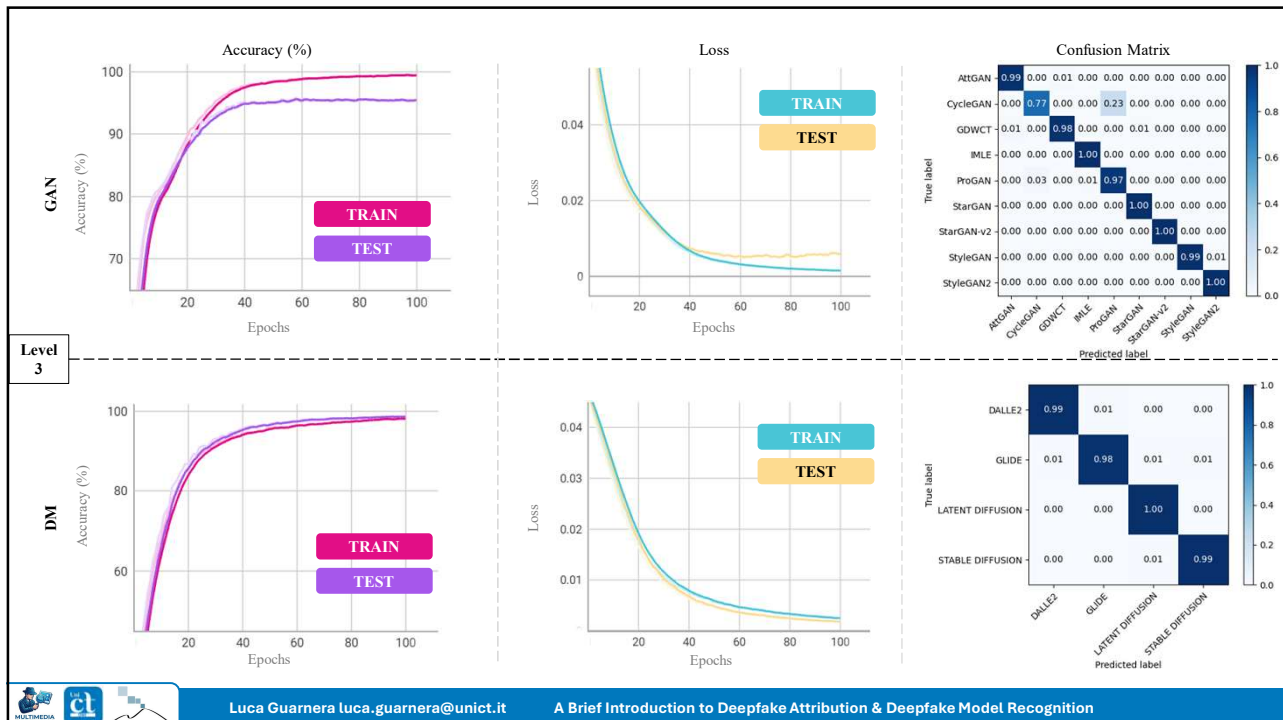
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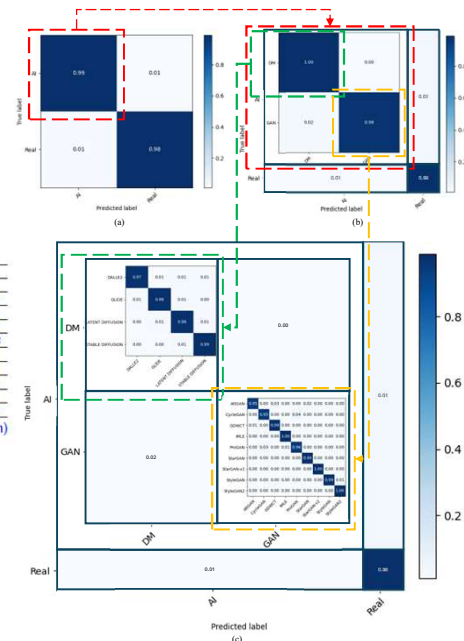


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Experimental Results

	Level 1				Level 2				Level 3 - DM				Level 3 - GAN				Overall Framework			
	Prec	Rec	F1	ACC	Prec	Rec	F1	ACC	Prec	Rec	F1	ACC	Prec	Rec	F1	ACC	Prec	Rec	F1	ACC
ResNET-18	0.84	0.98	0.89	0.965	0.95	0.98	0.96	0.9685	0.97	0.97	0.97	0.9716	0.98	0.98	0.98	0.9842	0.97	0.98	0.98	0.9686
ResNET-34	0.88	0.98	0.92	0.9761	0.95	0.97	0.96	0.9629	0.99	0.99	0.99	0.9898	0.98	0.98	0.98	0.9831	0.98	0.98	0.98	0.9621
ResNET-50	0.9	0.98	0.94	0.9818	0.94	0.97	0.95	0.9593	0.99	0.99	0.99	0.986	0.97	0.97	0.97	0.9771	0.97	0.98	0.97	0.9601
ResNET-101	0.94	0.99	0.96	0.9893	0.98	0.99	0.98	0.9845	0.99	0.99	0.99	0.9937	0.97	0.97	0.97	0.9701	0.97	0.99	0.98	0.9782
ResNET-101	0.86	0.98	0.91	0.9729	0.93	0.96	0.94	0.9481	0.97	0.97	0.97	0.9707	0.99	0.99	0.99	0.9925	0.98	0.98	0.98	0.952
DenseNET-121	0.88	0.99	0.92	0.9768	0.96	0.98	0.97	0.9751	0.99	0.99	0.99	0.9861	0.99	0.99	0.99	0.9893	0.99	0.99	0.99	0.972
EfficientNET-b4	0.7	0.93	0.76	0.8996	0.91	0.94	0.92	0.9317	0.79	0.78	0.78	0.7793	0.91	0.92	0.91	0.9141	0.9	0.91	0.9	0.8794
EfficientNET-b7	0.84	0.98	0.89	0.9661	0.91	0.95	0.93	0.935	0.93	0.92	0.92	0.923	0.98	0.96	0.96	0.9808	0.97	0.93	0.95	0.9341
ViT-B16	0.95	0.98	0.96	0.9904	0.92	0.96	0.94	0.945	0.98	0.98	0.98	0.9835	0.98	0.98	0.98	0.9794	0.97	0.98	0.98	0.9533

Table 2. Comparison of average precision, recall, f1-score and accuracy values obtained among all involved architectures (first column) on all levels of the proposed hierarchical approach.



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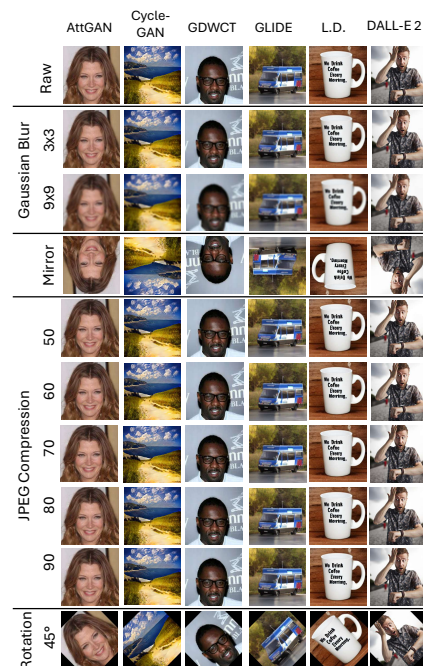
Comparison with SOTA approaches

	Level 1	Level 3	
	Real Vs AI	GANs	DMs
AutoGAN [46]	0,685	0,803	—
Fakespotter [16]	0,7422	0,9532	—
EM [21]	0,8657	0,9502	—
DCT[18]	0,8720	0,9589	—
Wang et al. [44]	0,7854	0,9732	—
DE-FAKE [41]	0,9052	—	0,9345
Our	0,9893	0,9701	0,9937

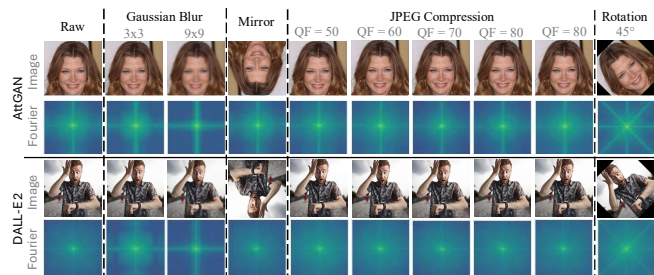
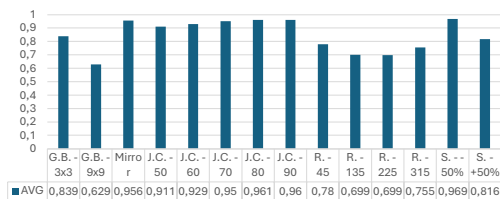
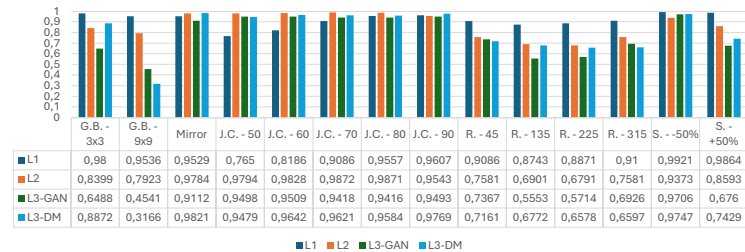
Table 3. Comparison with state of the art approaches. Classification accuracy value is reported. — are reported where the method can not handle the corresponding task.



Robustness Attacks



Robustness Attacks



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Generalization

- COCOFake [1]: among the most recent multimodal dataset on deepfakes, it is composed of images optimally manipulated through the diffusion model Stable Diffusion, starting from COCO dataset.
- FaceForensics++ (FF++) [2]: a dataset of deepfake videos.

	ACC	L1	L2	L3-DM	Overall
COCOFake [5]	0,9197	0,9302	0,7587	0,8538	
FF++ [39]	0,94	-	-	-	

Table 4. Generalization test performed on the COCOFake [5] and FaceForensics++ (FF++) [39] datasets. The classification accuracy value (ACC) is reported for each involved experiment.

1. Roberto Amoroso, Davide Morelli, Marcella Cornia, Lorenzo Baraldi, Alberto Del Bimbo, and Rita Cucchiara. 2023. Parents and Children: Distinguishing Multimodal Deepfakes from Natural Images. *arXiv preprint arXiv:2304.00500* (2023).
2. Andreas Rossler, Davide Cozzolino, Luisa Verdoliva, Christian Riess, Justus Thies, and Matthias Nießner. 2019. FaceForensics++: Learning to Detect Manipulated Facial Images. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 1–11.



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On the Exploitation of Deepfake Model Recognition



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Deepfake Model Recognition Task

Deepfake Model Recognition task

A fundamental element of all GAN architectures is their **latent space**!

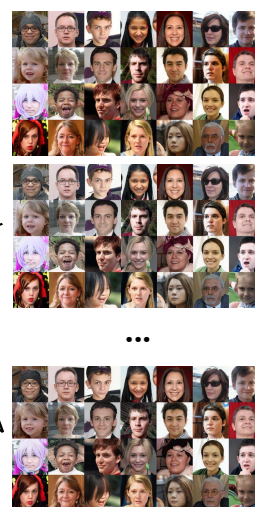
GENERATIVE
ARCHITECTURE

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Model-1

Model-2

Model-N



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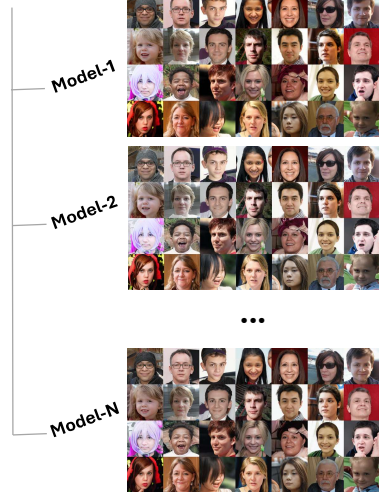
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Deepfake Model Recognition Task

The latent spaces of **50 slightly different StyleGAN-2^[A] models** were explored in depth.

[A] Tero Karras, Samuli Laine, Miika Aittala, Janne Hellsten, Jaakko Lehtinen, and Timo Aila. **Analyzing and improving the image quality of stylegan**. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 8110–8119, 2020.

StyleGAN-2



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StyleGAN 2 - generated images



[A] Tero Karras, Samuli Laine, Miika Aittala, Janne Hellsten, Jaakko Lehtinen, and Timo Aila. **Analyzing and improving the image quality of stylegan**. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 8110–8119, 2020.



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State of the art

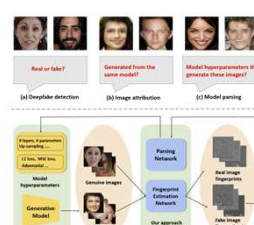


Fig. 1: Top: Three increasingly difficult tasks: (a) *deepfake detection* classifies an image as genuine or fake; (b) *image attribution* predicts which of a closed set of GMs generated a fake image; and (c) *model parsing*, proposed here, infers hyperparameters of the GM used to generate an image, for those models unseen during training. Bottom: We present a framework for model parsing, which can also be applied to simpler tasks of deepfake detection and image attribution.

Reverse Engineering of Generative Models: Inferring Model Hyperparameters from Generated Images

Vishal Asnani, Xi Yin, Tal Hassner, Xiaoming Liu

Abstract—State-of-the-art (SOTA) Generative Models (GMs) can synthesize photo-realistic images that are hard for humans to distinguish from genuine photos. Identifying and understanding manipulated media are crucial to mitigate the social concerns on the potential misuse of GMs. We propose to perform reverse engineering of GMs to infer model hyperparameters from the images generated by these models. We define a novel problem, “model parsing”, as estimating GM network architectures and training loss functions by examining their generated images—a task seemingly impossible for human beings. To tackle this problem, we propose a framework with two components: a Fingerprint Estimation Network (FEN), which estimates a GM fingerprint from a generated image by training with four constraints to encourage the fingerprint to have desired properties, and a Parsing Network (PN), which predicts network architecture and loss functions from the estimated fingerprints. To evaluate our approach, we collect a fake image dataset with 100K images generated by 110 different GMs. Extensive experiments show encouraging results in parsing the hyperparameters of the unseen models. Finally, our fingerprint estimation can be leveraged for deepfake detection and image attribution, as we show by reporting SOTA results on both the deepfake detection (Celeb-DF) and image attribution benchmarks.

Index Terms—Reverse Engineering, Fingerprint Estimation, Generative Models, Deepfake Detection, Image Attribution

1 INTRODUCTION

Image generation techniques have improved significantly in recent years especially after the breakthrough of Generative Adversarial Networks (GANs) [1]. Many Generative Models (GMs), including both GAN and Variational Autoencoder (VAE) [2], [3], [4], [5], [6], [7], [8], can generate photo-realistic images that are hard for human to distinguish from genuine photos. This photo-



Fig. 2: Example images generated by all 110 GMs in our collected dataset (one image per model).

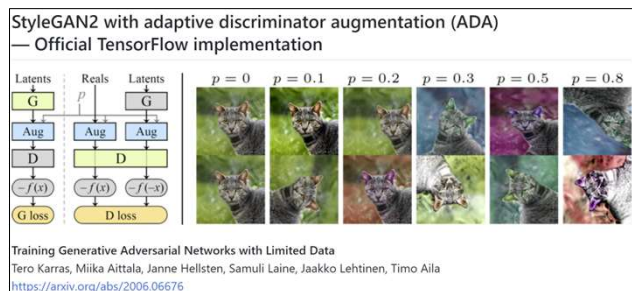
Paper: <https://arxiv.org/pdf/2106.07873.pdf>

Vishal Asnani, Xi Yin, Tal Hassner, and Xiaoming Liu. **Reverse engineering of generative models: Inferring model hyperparameters from generated images.** arXiv preprint arXiv:2106.07873, 2021.

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StyleGAN2-ADA

- The **StyleGAN2-ADA^[B]** implementation easily allows to **freeze the weights/parameters** of a user-defined **number of layers in the “Discriminator”** during the training operations.
- In order to have this freezing ability also available for the layers of the **“Generator”**, the original implementation of **StyleGAN2-ADA** was customized.



Github repo: <https://github.com/NVlabs/stylegan2-ada>
Paper: <https://arxiv.org/abs/2006.06676>

[B] Tero Karras, Miika Aittala, Janne Hellsten, Samuli Laine, Jaakko Lehtinen, and Timo Aila. **Training generative adversarial networks with limited data.** In H. Larochelle, M. Ranzato, R. Hadsell, M. F. Balcan, and H. Lin, editors, *Advances in Neural Information Processing Systems*, volume 33, pages 12104–12114. Curran Associates, Inc., 2020.

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The Models

- Let M_{ST2} be the well-known pre-trained StyleGAN2 model (<https://github.com/NVlabs/stylegan2>)
- Fine-tune operations** were carried out by varying only the **hyper-parameters**:
 $k\text{-img} \in \{1, 2, \dots, 10\}$ and $p \in [0, 1]$
 which describe **training-set augmentation probability**.
- All possible combinations of pairs $(\bar{k}, \bar{p})$ with $\bar{k} \in k\text{-img}$ and $\bar{p} \in p$ were considered (examples (1, 0.1), (1, 0.2), ... (10, 0.9), (10, 1.0)) to obtain a total of **100 StyleGAN2-ADA fine-tuned models** ($M_{ST2-ADA}$).
- For each $M_{ST2-ADA}$, **1.000 images were generated** for a total of **100.000 images**.

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Images generated by fine-tuned StyleGAN2-ADA models



Figure 2. Generated image examples from the slightly different models employed in the experiments.

www.capturethefake.com

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The Models

Let $M_i(W)$ be the i -th model of $M_{ST2-ADA}$ uniquely defined by its set of parameters (weights)

$$W = (W_G, W_D)$$

with $W_G = \{w_{G1}, w_{G2}, \dots, w_{Gn}\}$ and $W_D = \{w_{D1}, w_{D2}, \dots, w_{Dm}\}$; n and m are the total number of layers in G and D respectively

Specifically, in the fine-tuning procedure, as far as the "Generator" and the "Discriminator", **only the layers reported in the table were not frozen.**

Generator	Discriminator
synthesis.b1024.conv0	b4.mbstd
synthesis.b1024.conv1	b4.conv
synthesis.b1024.torgb	b4.fc
synthesis.b1024:0	b4.out
synthesis.b1024:1	

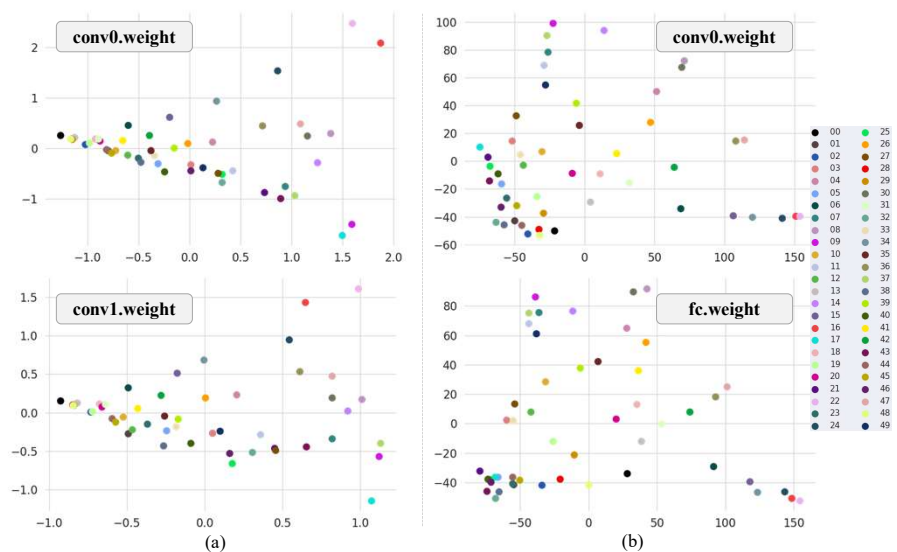
Table 1. StyleGAN2 Generator G and Discriminator D structure. The first column describes layer $b1024$ in the synthesis block in G ; the second column represents the structure of layer $b4$ in D . These blocks represent the layers that were re-trained during the fine-tuning procedure



The Models

Figure 3. 2D PCA of the parameters extracted from the last layer of the Generator (a) and the Discriminator (b) of each of the 50 models.

Some weights of only the **non-frozen layer** of the Generator and Discriminator are shown



The Images

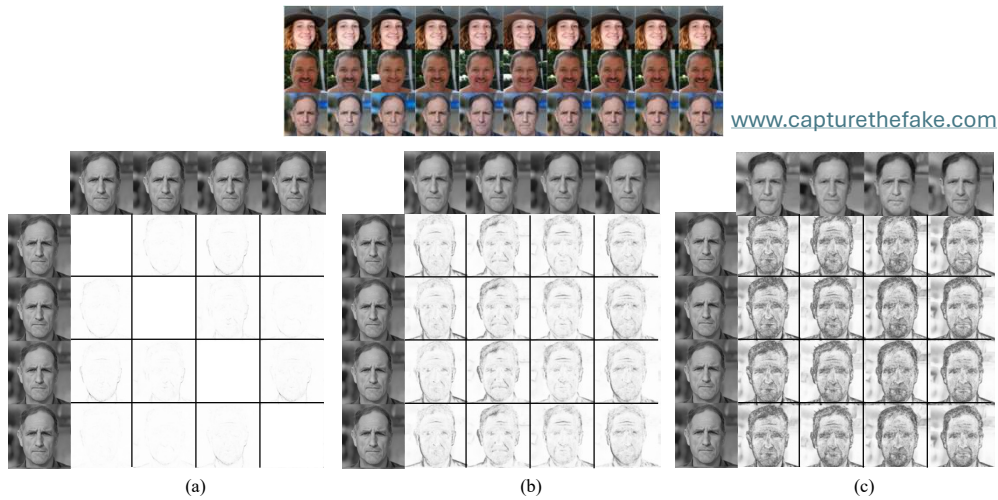


Figure 4. SSIM maps comparisons between gray-scale images. (a) shows samples of extremely similar images; (b) shows samples where the differences are related mainly to the smoothing effects. (c) shows examples where the difference is more evident and related to misalignments



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Training and Test data definition

- 100 different models were created.
- Only 50 (M_{1-50}) of the 100 $M_i(W)$ models were randomly selected for the training procedures.
- For each of the selected model, 1.000 images were generated.
- Testing was carried out on different sets of images.

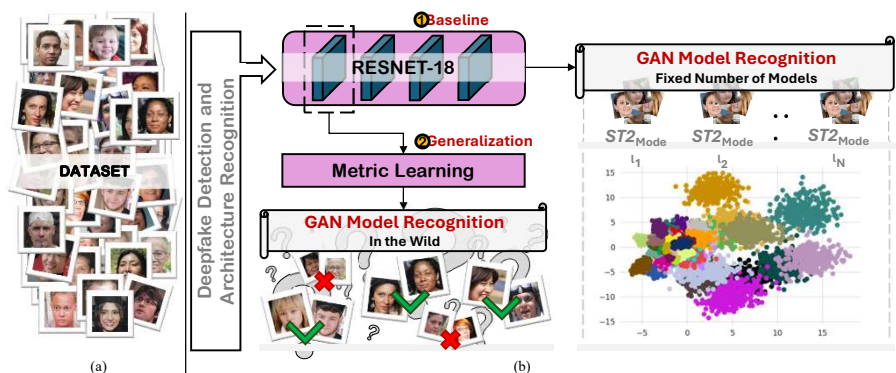


Figure 1. Schematic representation of the robust processing pipeline proposed for Deepfake Model Recognition. At first, a dataset has to be created (a) by fine-tuning different models and thus generating images. In general, an efficient Deepfake detector/classifier could be employed at first from those available in literature. Then, an encoder (b) is trained for the recognition task on the a-priori known models. The obtained feature vectors on a two-dimensional space shows the discriminative power of features encoded by the baseline encoder. Finally, the first layer of the trained encoder is exploited in a metric learning approach.



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Deepfake Model Recognition Task

- The objective of a **Deepfake Model Recognition task** is to assess whether **two images were generated by the same model** by defining a metric **L** . Formally

$$L(F(I_a^i), F(I_b^j)) \in [-1, 1] \quad (1)$$

- where $I_a^i \in I_{M_i(W)}$ and $I_b^j \in I_{M_j(W)}$ are two images generated respectively by $M_i(W)$ and $M_j(W)$ models ($M_i \in M_{ST2-ADA}$ and $M_j \in M_{ST2-ADA}$), and **$F(\cdot)$** is a function for fingerprint extraction from a given input image.

Note that this task is completely different from those of **Deepfake Detection** and **Architecture Recognition**



Deepfake Model Recognition Task

- To **define and to train an encoder (F)**, able to discriminate 50 different classes (M_{1-50}) and demonstrate its effectiveness by measuring the accuracy even in **noisy scenarios**;
- To **employ the trained classifier**, as an embedding method (F) that is able to exploit the latent spaces, even for images generated by models **never seen during the training phase** (M_{51-100});
- To **define and to learn a robust metric L** to compare two images I and assess whether they were generated by the **same model**



Experimental Results

A-priori known models classification results

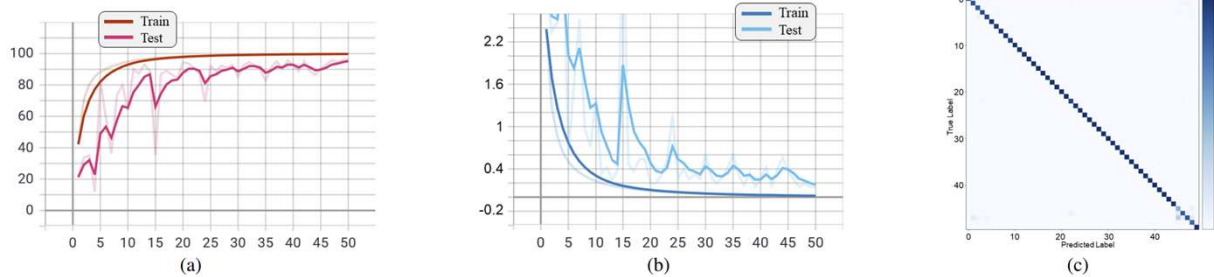


Figure 6. Training accuracy (a) and loss (b) through 50 epochs on raw-images. The confusion matrix in (c) shows classification results obtained by the best classification model, over 50 classes.

Average classification accuracy value of 96.24% was obtained.



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Experimental Results

A-priori known models classification results

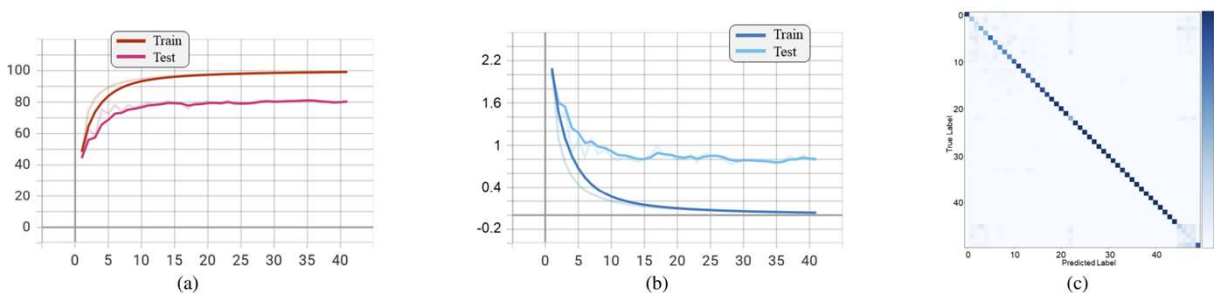


Figure 7. Training accuracy (a) and loss (b) through 41 epochs on noisy images. The confusion matrix in (c) shows classification results obtained by the best classification model, over 50 classes.

Average classification accuracy value of 81.37 % was obtained.



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Learning a metric



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Learning a metric

2012 IEEE 12th International Conference on Data Mining

Metric Learning From Relative Comparisons by Minimizing Squared Residual

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Abstract—Recent studies [1]–[5] have suggested using constraints in the form of relative distance comparisons to represent domain knowledge: $d(a, b) < d(c, d)$ where $d(\cdot, \cdot)$ is the distance function and a, b, c, d are data objects. Such constraints are readily available in many problems where pairwise constraints are not natural to obtain. In this paper we consider the problem of learning a Mahalanobis distance metric from supervision in the form of relative distance comparisons. We propose a simple, yet effective, algorithm that minimizes a convex objective function corresponding to the sum of squared residuals of constraints. We also extend our model and algorithm to promote sparsity in the learned metric matrix. Experimental results suggest that our method consistently outperforms existing methods in terms of clustering accuracy. Furthermore, the sparsity extension leads to more stable estimation when the dimension is high and only a small amount of supervision is given.

Keywords—metric learning; Mahalanobis metric; relative comparisons;

I. INTRODUCTION

third object, but we cannot determine an absolute closeness measure.

Knowledge of relative comparisons are readily available in many real world problems. One example is the ranking problem in search engine (ranking search results or relevance based advertisements). Query and user-click logs can effectively provide knowledge in relative comparisons but not easily expressed in pairwise constraints [1]. A metric learned from these comparisons can reveal feature importance and feature-feature interactions. Thus it can provide valuable guidance in feature-engineering of ranking problems (e.g., to design cross-features). Another example is clustering with structural knowledge as supervision. We have previously shown that paired relative comparisons can be used to express structural knowledge unambiguously [5]. Existing structural knowledge sources such as taxonomical databases can be effectively translated into a minimum set of relative comparisons. For in-

Eric Yi Liu, Zhishan Guo, Xiang Zhang, Vladimir Jojic, and Wei Wang. **Metric learning from relative comparisons by minimizing squared residual**. In 2012 IEEE 12th International Conference on Data Mining, pages 978–983. IEEE, 2012.



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Experimental Results

Unknown model classification results

The **metric L** was trained on **a-priori known models** (M_{1-50} of the dataset) by means of F and several test was performed by considering:

- Images generated by a-priori known models M_{1-50} ;
- Images generated by unknown models never used for any training operation M_{51-100} ;
- The union of two previous sets and with the noises attacks
- Images generated by StyleGAN-1
- Images generated by two Stylegan-1 models.

	1-NN	3-NN	5-NN
A-priori known	95.08%	95.74%	95.74%
Unknown models	92.31%	92.55%	94.33%
Known + unknown	93.70%	94.24%	95.06%
Noise-augmented	75.66%	77.00%	79.66%
ST1-M1 vs. StyleGAN2	99.98%	99.98%	99.98%
ST1-M1 vs. ST1-M2	84.50%	88.20%	92.00%

Table 2. Results of the learnt metric L by means of employing it in a k-NN classification test. Values reported are the average accuracies obtained on 5-fold cross validation tests employing 20% of data as reference and the remaining for classification. ST1-M1 stands for the first model of StyleGAN1.

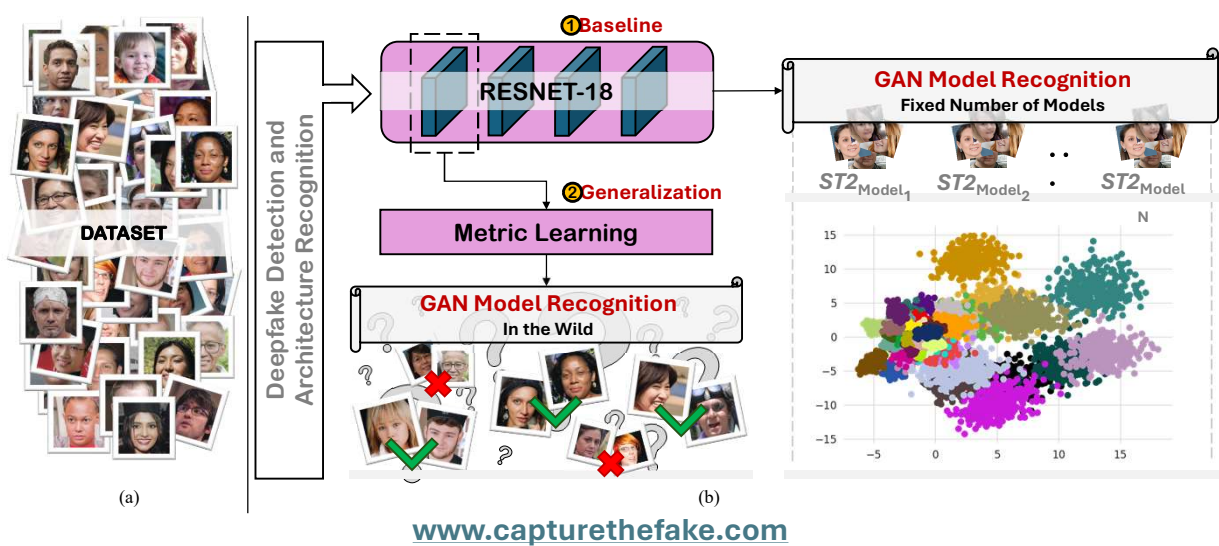


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Conclusion



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